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V12Bio

Virudhunagar District

Common Quarterly Examination - September 2026

Standard 12

Time Allowed: 3.00 Hours **BIOLOGY** Maximum Marks: 70

Part - I

Bio-Botany (35 marks)

I. Choose the correct answer: 8 x 1 = 8

1. Match the following.

I)	External fertilization	- i)	Pollen grain
II)	Androecium	- ii)	Anther wall
III)	Male gametophyte	- iii)	Algae
IV)	Primary parietal layer	- iv)	Stamens

a) I-iv, II-i, III-iii, IV-iii b) I-iii, II-iv, III-i, IV-ii
c) I-iii, II-iv, III-ii, IV-ii d) I-iii, II-i, III-iv, IV-ii

2. ITransmitting tissue is found in
a) microphylar region of ovule b) pollen tube wall
c) stylar region of gynorcium d) integument

3. Gene which suppresses other genes activity but does not lie on the same locus is called as
a) epistalic b) supplement only c) hypostatic d) codominant

4. Incomplete dominance was reported in
a) mirabilis b) wheat c) tobacco d) cucurbita

5. ECORI cleaves DNA at
a) AGGGTT b) GTATATC c) GAATTC d) TATAGC

6. Changing the codon AGC to AGA represents
a) missense mutation b) nonsense mutation
c) frameshift mutation d) deletion mutation

7. Pedogenesis refers to
a) fossils b) water c) population d) soil

8. Who proposed the concept of totipotency?
a) Karl Ereky b) Gottlieb Haberlandt c) Edward Jenner d) Ernst Hoppe

II. Answer any 4 questions. 4 x 2 = 8

9. What is reproduction?
10. What is stomium?
11. Write short notes on Sonara 64
12. What are the materials used to grow microorganism like spirulina?
13. Differentiate Primary metabolites and secondary metabolites.
14. Distinguish habitat and niche.

III. Answer any 3 questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. Differentiate grafting and layering.
16. What is meant by cytoplasmic inheritance?
17. What is Albedo effect and write their effects.
18. Write any three application of Biotechnology.
19. Give examples of three secondary metabolites metabolites produced by plants resources.

IV. Answer all the questions. 2 x 5 = 10

20. a) With a suitable diagram, explain the structure of an ovule.

(OR)

b) Explain incomplete dominance with example.

21. a) Mention the name of man made cereal. How it is formed?

(OR)

b) Write the benefits and risk of genetically modified foods.

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Part - II
Bio-Zoology (35 marks)

I. Choose the correct answer: 8 x 1 = 8

1. In which type parthenogentic egg may develop into individuals of any sex
   a) Arrhenotoky b) Thelytoky c) Amphitoky d) Both (a) and (b)
2. Colostrum is rich in
   a) IgE b) IgA c) IgD d) IgM
3. A contraceptive pill prevents ovulation by
   a) blocking follopian tube b) inhibiting release of FSH and LH
   c) stimulating release of FSH and LH d) degeneration of released ovum
4. Down's syndrome is also referred to as
   a) 21-Trisomy b) 13-Trisomyc) 18-Trisomy d) None of these
5. A mRNA molecule is produced by
   a) Replication b) Translation c) Duplication d) Transcription
6. Which period was called "Age of Fishes"?
   a) Permian b) Triassic c) Devonian d) Ordovician
7. Choose the correctly match pair.
   a) LSD - Narcotic
   b) Amphetamines - Stimulant
   c) Heroin - Psychotropic
   d) Benzodiazepine - Painkiller
8. Match the pathogens with respective diseases caused by them and select the correct match using the codes given below.
   A) Leishmania donavani - i) Amoebiasis
   B) Wuchereria bancrofti - ii) Kala-azar
   C) Trypanosoma gambiense - iii) Sleeping sickness
   D) Entamoeba histolytica - iv) Filariasis
   a) A-ii, B-iv, C-iii, D-i b) A-ii, B-iv, C-i, D-iii
   c) A-iii, B-i, C-ii, D-iv d) A-i, B-iv, C-iii, D-ii

II. Answer any 4 questions. 4 x 2 = 8

9. What is parthenogenesis? Give two examples from animals.
10. How is polyspermy avoided in humans?
11. Why the human genome project is called a mega project?
12. What are interferons? Mention their role.
13. How does Neanderthal man differ from the modern man in appearance?
14. Explain Surrogacy.

III. Answer any 3 questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. List the common withdrawal symptoms of drugs and alcohol abuse.
16. Differentiate between divergent evolution and convergent evolution with one example for each.
17. What are the three structural differences between RNA and DNA.
18. Explain the mode of sex determination in honey bees.
19. Write short notes on the following :
    a) ZIFT b) ICSI

IV. Answer all the questions. 2 x 5 = 10

20. a) Explain the various phases of the menstrual cycle.

(OR)

    b) How is sex determined in human beings?

21. a) Explain experimental approach to the origin of life.

(OR)

    b) Explain briefly the structure of immunoglobulin with suitable diagram.

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